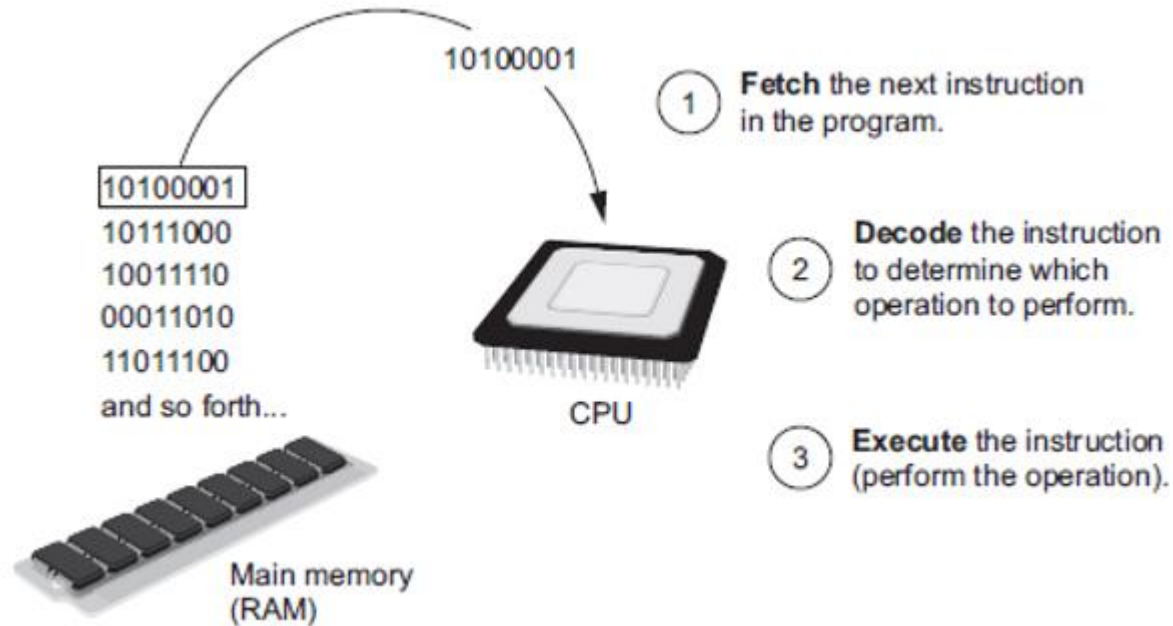




How Programs Work

- **Program must be copied from secondary memory to RAM each time CPU executes it**
- **CPU executes program in cycle:**
 - **Fetch**: read the next instruction from memory into CPU
 - **Decode**: CPU decodes fetched instruction to determine which operation to perform
 - **Execute**: perform the operation

How Programs Work (cont'd)





From Machine to Assembly Language

- **Impractical for people to write in machine language**
- **Assembly language: uses short words (mnemonics) for instructions instead of binary numbers**
 - Easier for programmers to work with
- **Assembler: translates assembly language to machine language for execution by CPU**

From Machine to Assembly Language (cont.)

Assembly Language

```
LDA A   ACIA
ASR A
BCC     INCH
LDA A   ACIA+1
```



Assembler

Machine Language

```
0000 1001 1100 0110 1010 1111 0101 1000
1010 1111 0101 1000 0000 1001 1100 0110
1100 0110 1010 1111 0101 1000 0000 1001
0101 1000 0000 1001 1100 0110 1010 1111
```

High-Level Languages

❧ **Low-level language**: close in nature to machine language

- Example: assembly language

❧ **High-Level language**: allows simple creation of powerful and complex programs

- No need to know how CPU works or write large number of instructions
- More intuitive to understand

Compilers and Interpreters

- Programs written in high-level languages must be translated into machine language to be executed
- Source code: statements written by programmer
- Compiler: translates high-level language program into separate machine language program
 - Machine language program can be executed at any time
 - Example: C/C++ is a compiled language

Compilers and Interpreters (cont'd)

- 1 The compiler is used to translate the high-level language program to a machine language program.

High-level language program

```
print ("Hello  
Earthling")  
  
and so forth...
```



Compiler



Machine language program

```
10100001  
10111000  
10011110  
and so forth...
```

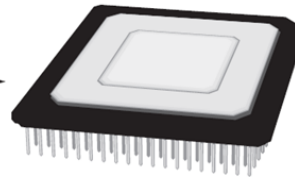
- 2 The machine language program can be executed at any time, without using the compiler.

Machine language program

```
10100001  
10111000  
10011110  
and so forth...
```

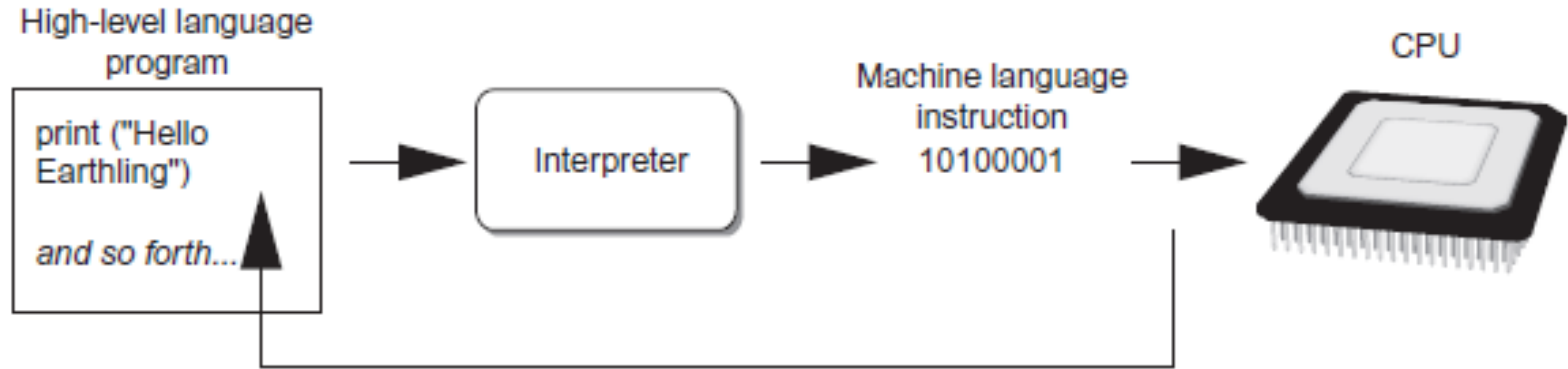


CPU



- **Interpreter: translates and executes instructions in high-level language program**
 - Interprets one instruction at a time
 - No separate machine language program
 - Used by Python language

Compilers and Interpreters (cont'd)



The interpreter translates each high-level instruction to its equivalent machine language instructions and immediately executes them.

This process is repeated for each high-level instruction.



Python

- **A general-purpose programming language**
 - Can be used to write code for any programming task
- **An interpreted language (vs. compiled language)**
 - Code is translated by an interpreter one statement at a time
 - In a compiled language, the entire source code is translated then executed altogether
- **An object-oriented language**
 - Data in Python are objects created from classes
 - A class is a type that defines objects of the same kind with common properties and methods to manipulate them
- **Installing Python:** <https://www.anaconda.com/download>